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Designing and analysing powerful experiments: practical tips for applied researchers

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Abstract

This paper offers practical advice on how to improve statistical power in randomised experiments through choices and actions researchers can take at the design, implementation and analysis stages. At the design stage, the choice of estimand, choice of treatment, and decisions that affect the residual variance and intra-cluster correlation can all affect power for a given sample size. At the implementation stage, researchers can boost power through increasing compliance with treatment, reducing attrition and improving outcome measurement. At the analysis stage, power can be increased through using different test statistics or estimands, through the choice of control variables, and through incorporating informative priors in a Bayesian analysis. A key message is that it does not make sense to talk of 'the' power of an experiment. A study can be well powered for one outcome or estimand but not others, and a fixed sample size can yield very different levels of power depending on researcher decisions.

KEYWORDS
statistical power, randomised experiment, experimental design, take-up

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1 | INTRODUCTION

The last two decades have seen rapid growth in the use by economists of randomised experiments to answer policy and research questions. As of 24 March 2025, there were more than 10,000 studies registered in the American Economic Association's RCT Registry, with locations in 170 countries. Typically, these studies aim to test the impact of one or more policy interventions or treatments, and conduct hypothesis tests to determine whether or not the treatment had a significant impact. The ability of these experiments to detect a treatment effect depends on their statistical power, which is the probability of rejecting the null hypothesis of no effect when this null is false. But many of

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these experiments may be underpowered. Ioannidis, Stanley and Doucouliagos (2017) claim that half of the research areas in economics have nearly 90 per cent of their results underpowered, although they do not distinguish between experimental and non-experimental estimates.[1] DellaVigna and Linos (2022) find that one reason for the much smaller effects of nudge treatments when delivered at scale is the low power of many academic experiments, with publication bias exacerbating this effect as low-powered studies that fail to find an effect are less likely to get published. Conducting an underpowered study can waste researcher time and the resources of funders, and lead to inconclusive results that fail to inform policy or that lead to the wrong policy conclusions. For example, McKenzie (2021) suggests that the conclusion that business training does not work was a result of most studies in the literature being underpowered to detect the 5–10 per cent improvement that meta-analysis reveals. While increasing the sample size is the obvious solution, Faridani (2024) calculates that power in economics experiments published in top journals would only increase by 7.2 percentage points on average if sample sizes were to be doubled.

This paper provides practical guidance of approaches other than simply increasing sample size that can be used to design and analyse powerful experiments. Although my focus and experience are based on field experiments, many of the details will also apply to lab experiments and to non-experimental evaluations.[2] Most discussions of power calculations take the estimand, treatment, outcome of interest, variance and intra-cluster correlation as given, with the focus then on either calculating the minimum detectable effect (MDE) in a given sample N or calculating the sample size N needed to achieve a desired MDE. In contrast, this paper argues that choices made in the design, implementation and analysis stages of an experiment can change all of these inputs, and thereby improve statistical power. I use an example of an experiment in Colombia designed to boost firm exports to illustrate some of these points.

At the design stage, I discuss four types of choices researchers can make to improve power. The first is changing the population of interest, and hence estimand, by deciding who should not be in the study, screening out units that are likely to attrit, are unlikely to comply with treatment or are outliers. Second, the same experiment can be well powered for some outcomes and not others. Choosing outcomes that are closer to the intervention in the causal chain can improve power. Decisions over whether to use a continuous, winsorised or binary measure can also greatly influence power. Power can be improved by the use of multiple follow-up measures, especially for outcomes that are not highly autocorrelated, by changing the estimand to an average impact pooled across rounds. In clustered experiments, the intra-cluster correlation will vary across outcomes, and power can be affected by variability in cluster size.

At the implementation stage, power can be increased by actions to boost take-up rates and reduce non-compliance, including the use of incentives, reminders and waiting lists. Survey efforts to reduce attrition increase the effective sample size and power. Careful attention to measurement can reduce the residual variance and make it easier to detect treatment effects.

At the analysis stage, the choice of test statistic can affect power. I discuss when and whether to use one-sided instead of two-sided tests, the use of estimands that put more weight on units and outcomes that are affected more by the treatment, and tests that focus on determining whether there was any effect rather than what that effect was. Power can be increased through soaking up residual variation through including additional control variables. The lagged dependent variable in ANCOVA (analysis of covariance) estimation is the most important here, especially when outcomes are highly autocorrelated, whereas in practice few other control variables offer much in the way of power gains in most settings. Finally, incorporating outside information in the form of informative priors can boost power through Bayesian analysis. Taken together, this set of actions can greatly boost power for a given sample size.

[1] This problem is not unique to economics: Arel-Bundock et al. (2024) estimate that median analysis in quantitative political science 'has about 10% power, and only about 1 in 10 tests have at least 80% power to detect the consensus effects reported in the literature'.

[2] Of course, non-experimental approaches also need to worry more about internal validity, and some of the recommendations here to increase power may affect bias in those studies. There are also additional issues around power that can arise in non-experimental studies since often it is the case that not all units contribute to identification. See McKenzie (2011 and 2016) for examples.

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This paper contributes to a broader literature on applied issues in experimental design in economics. Excellent general introductions are provided by Duflo, Glennerster and Kremer (2007) and the textbook of Glennerster and Takavarasha (2013). In previous work, I have examined issues around how to randomise (Bruhn and McKenzie, 2009), how to use multiple time periods to improve power (McKenzie, 2012) and whether poverty reduction policies are useful at all if they require power calculations to tell whether they have worked (McKenzie, 2020). There is also a useful literature on how to do power calculations once we deviate from the standard textbook cases (e.g. Burlig, Preonas and Woerman, 2020; Blair, Coppock and Humphreys, 2023; McConnell and Vera-Hernández, 2025). This paper builds on this work to offer specific, practical actions researchers can take to improve power.

2 | THE BASIC SET-UP: STATISTICAL POWER FOR DETECTING A TREATMENT EFFECT IN AN EXPERIMENT

Consider designing an experiment to test the impact of some binary treatment T on a given outcome Y, which can be either continuous or binary. The researcher then needs to specify the *estimand*, which is the specific parameter or quantity of interest. In many randomised experiments, this is the intention-to-treat (ITT) effect, which is the average effect in the population A of interest of being assigned to the treatment rather than the control:

$$\beta = E\left(Y_i|T_i = 1, i \in A\right) - E\left(Y_i|T_i = 0, i \in A\right). \tag{1}$$

In an individually randomised experiment, a sample N from the population A is selected, and then a proportion P are randomly assigned to the treatment group, and proportion 1–P to the control group. Then the OLS (ordinary least squares) estimator $\hat{\beta}$ of this treatment effect is obtained by running a simple regression:[3]

$$Y_i = \alpha + \beta T_i + \varepsilon_i. \tag{2}$$

The variance of the OLS estimate of the treatment effect β is then given by

$$Var\left(\hat{\beta}\right) = \frac{1}{P\left(1-P\right)}\frac{\sigma^2}{N}, \tag{3}$$

where σ^2 is the variance of the residual term ε_i. Then a test of the null hypothesis that $\beta = 0$ – i.e. that there is no average effect of being assigned to treatment – will reject the null if

$$\frac{\left|\hat{\beta}\right|}{\sqrt{Var\left(\hat{\beta}\right)}} > t_{\alpha/2}, \tag{4}$$

where α is the significance level (typically 5 per cent) and $t_{\alpha/2}$ is the critical value from the standard normal distribution (1.96 when $\alpha = 0.05$). The statistical power of this test is the probability that this test will reject the null when the null is false, which is 1 minus the probability κ of a type II error

[3] When Y is not continuous, researchers may instead wish to move away from a linear model to use a logistic model (for binary outcomes) or Poisson or negative binomial model (for count data). The variance term and power calculation will differ slightly in these cases (see, for example, McConnell and Vera-Hernández (2025) for the binary case). However, in practice the differences are usually small, and since economists tend to prefer using linear models, I focus on this case.

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(failing to reject the null when the null is false). We can either fix power (typically at 80 per cent) and use this to derive the minimum detectable effect (MDE) size:

$$MDE = \left(t_{\alpha/2} + t_{1-\kappa}\right)\sqrt{\frac{\sigma^2}{P(1-P)N}} \tag{5}$$

or rearrange this (see, for example, Glennerster and Takavarasha (2013, pp. 269–70)) to obtain the formula for statistical power:

$$1-\kappa = \Phi\left(\left[MDE \times \sqrt{P(1-P)\frac{N}{\sigma^2}}\right] - t_{\alpha/2}\right), \tag{6}$$

where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution. Since this is an increasing function, it shows that, all else equal, power will be higher the larger the sample size N, the smaller the residual variance σ^2, and when $P = \frac{1}{2}$ so that units are equally allocated to treatment and control. Note that there are settings where users may wish to deviate from a 50:50 allocation to treatment and control. The first is when the binding constraint is not the sample size N available but the budget available for including units into the experiment. Then if including treatment units is more costly than including control units, power can be maximised subject to the budget constraint by allocating relatively more units to control than treatment.[4]

A second, and less common, reason is when there is a strong prior that treatment may change the variance as well as the mean for the treatment group. With a binary outcome, the variance of Y is $\pi_T(1-\pi_T)$ where π_T is the mean of Y in treatment group T. As a result, if there is a treatment effect that changes the proportion with $Y = 1$ between groups, this also results in the two groups having different variances. However, under the null, the variances are the same, and under moderate effect sizes the difference in variances is typically very small provided we are not near the tails. So even in the binary case, I would assume equal variances for power calculations.

I see the strongest case for adjusting the proportions from 50:50 to be when it is expected that only a few people will benefit from treatment, and the outcome is one in which the control group and those who do not benefit from the treatment see little variation.[5] Then it may be a lot harder to estimate $E(Y_i|T_i = 1, i \in A)$ than $E(Y_i|T_i = 0, i \in A)$, and we may wish to allocate more units to treatment as a result. As a concrete example, suppose the treatment is a migration facilitation programme, the outcome is earnings from migration, and it is anticipated that (almost) no one in the control group will migrate and that only 10–15 per cent of the treatment group will migrate and work. Then if all the heterogeneity in outcome is coming from those who migrate, we may wish to place more units in treatment to yield a larger sample of migrants.

Researchers also commonly conduct cluster-randomised experiments, in which clusters such as schools, health clinics or villages are randomised to treatment and control. When the sample size N is divided into J clusters, with an identical number of units per cluster, n, then Glennerster and Takavarasha (2013) note that the MDE in equation (5) gets multiplied by an additional term known as the design effect:

$$D = \sqrt{1 + (n-1)\rho}, \tag{7}$$

[4] Duflo, Glennerster and Kremer (2007) show that the ratio of the proportion of subjects allocated to the treatment group to those allocated to the control should be equal to the square root of the ratio of their costs.

[5] Incomplete take-up is a sub-case of this, but I would only suggest putting more in the treatment group than in the control for take-up reasons if the sample size is relatively small and take-up rates are anticipated to be 20 per cent or lower. Even then, the main reason I would do this is to ensure I have enough observations that actually participate in treatment for looking at their characteristics to predict who complies, and for getting qualitative insights on how treatment has affected them, than for reasons of power.

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where ρ is the intra-cluster correlation. The higher ρ, the higher the MDE for a given power, or the lower the power for detecting a given MDE. There is an additional complication when the clusters are not of equal size. Eldridge, Ashby and Kerry (2006) note that when clusters are unequally sized, estimates from the smaller clusters will be less precise than those from larger clusters, but since there are diminishing gains from adding more units to the same cluster, overall power will decrease as the cluster sizes become more unbalanced. This is seen in the design effect term becoming

$$D = \sqrt{1 + \rho\left(n\left(CV^2 + 1\right) - 1\right)}, \tag{8}$$

where CV is the coefficient of variation in cluster size and n is now the average cluster size. If the clusters are equally sized, $CV = 0$, and equation (8) collapses to equation (7), but otherwise the MDE becomes larger or power smaller, the more unequal are cluster sizes.

Most of the textbook and theoretical analysis takes the estimand, treatment, outcome, variance and intra-cluster correlation as given, and sets the problem as one of choosing what sample size N to use. However, in practice, the sample size is often dictated by the policy or budget constraints, whereas choices made by researchers can change all of these, leading to more powerful experiments.

3 | A PRACTICAL EXAMPLE

To illustrate some of the practical issues that arise and ways in which the power of the study can be changed through design and analysis choices, I will use the case of an export improvement programme that was implemented by the government of Colombia and studied in Iacovone, McKenzie and Meager (2025). I use data from the replication package (Iacovone, McKenzie and Meager, 2024). This programme was launched by the government with the goal of improving the productivity and export capacity of participating firms. The treatment here consisted of consulting services intended to improve the management practices of firms. Firms applied to the programme, and 200 firms met the government's eligibility criteria. Figure 1 shows histograms of the baseline number of employees and value of exports of these 200 firms. They are very heterogeneous: mean (median) firm size is 73 (42) employees, with a standard deviation of 106 employees, and range from 2 to 750 employees. Half of the firms had 0 exports in the year before the programme, the mean was $338,000 and standard deviation more than $1 million.

4 | IMPROVING POWER AT THE DESIGN STAGE

Researchers can improve power at the design stage of an experiment through careful consideration of the choice of estimand, through the choice of treatment(s), and through design choices to reduce the variance and intra-cluster correlation.

4.1 | The choice of estimand

4.1.1 | Deciding who should not be in your study (changing A)

Equation (1) defined the estimand of interest β as the ITT for individuals in a specified population A. However, a given experiment may have significantly more power for detecting treatment effects in a subpopulation A_1 than in the full population A. In particular, there are three types of units that researchers may wish to screen out. The first is those for whom follow-up data are unlikely to be available at all, i.e. those who are more likely to attrit (since attrition reduces the effective sample

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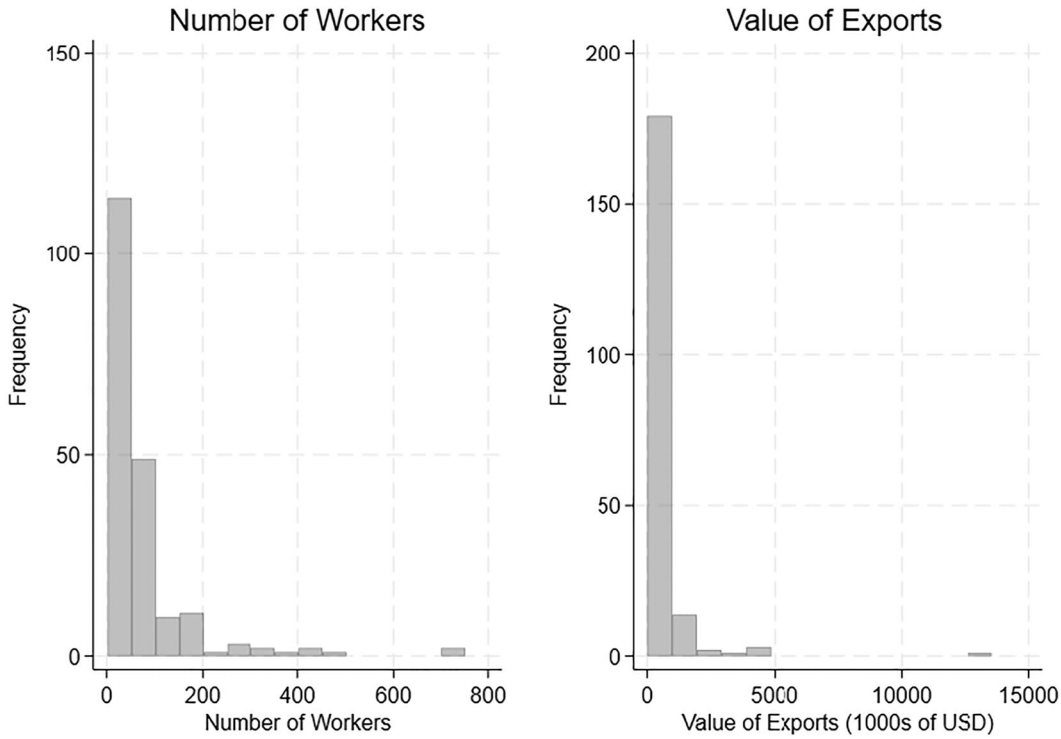


FIGURE 1 Baseline employees and exports of Colombian firms in export improvement experiment.

size). For example, Fafchamps et al. (2014) conducted an experiment to measure the impacts of cash and equipment grants on microenterprises in Ghana. They first conducted a screening survey to get microenterprises of their target type and size, and then two rounds of pre-treatment surveys, spaced one quarter apart. Only firms that responded to both survey rounds, and were still operating a business, were randomised into treatment and control. This eliminated 13 per cent of the baseline sample, leaving a sample more likely to answer surveys and not to attrit.

The second type of unit that researchers may wish to screen out is individuals that are not very likely to comply with treatment or be affected by it. For example, researchers wanting to test the impact of microfinance may wish to work with a population that applies for credit, or expresses interest in receiving credit, rather than a random sample of the population. However, there are cases where those who are the least interested in a programme may be the ones to benefit most from it (e.g. Sandvik et al., 2023), and so screening on interest may not always be a way to boost power.

The third type of exclusion that researchers may wish to make is to exclude units that are very different from everyone else in the population, i.e. to exclude outliers and make the sample more homogeneous. In our Colombian export project example, there are a few firms that are much larger and more successful at exports than anyone else. Mean outcomes will be dominated by them if they are included in the study. Here, dropping some firms and reducing N can actually increase power or reduce the MDE, because it reduces σ^2 by more than it reduces N, thereby reducing σ^2/N. For example, in the full sample of applicants, mean employment is 73 with a standard deviation of 106, while dropping the top 5 per cent of firms reduces the mean to 53 and standard deviation to 50. The power to detect a (very large) 20-worker increase from treatment with the full sample of 200 is then only 26.6 per cent, whereas it would be 78.7 per cent in the smaller sample of 190 firms that removed these outliers. An alternative to dropping these firms completely is to put them in their own stratum, so that the sample is randomised for the subpopulation of non-outliers A_1 as well as for the full sample

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A.[6] A further way to change A is to focus on a very homogeneous sample – so, for example, considering only farmers farming a single type of crop for an agricultural study, or firms all of a specified size working in the same industry for a firm study, or workers all trained in a single skill in a jobs study. The shocks these units experience will be more likely to be similar to one another and average out, making it easier to detect changes in the outcome coming from treatment.

The natural concern with changing the study population from A to A_1 is one of external validity. We only learn the treatment effect for the subset of firms that are less likely to attrit, more likely to comply with treatment and/or who are not outliers. But in practice I do not think we lose much here. Even if we were to include such units, we would find it difficult or impossible to learn treatment impacts for units that attrit. Learning that the ITT is close to zero because most people in the population do not take up treatment may be an interesting policy parameter, but does not help us learn the impact of the treatment for those who do take up treatment.[7] And if most firms have 40 or 50 workers, and only two have 750 workers, we cannot estimate the impact of treatment for that subgroup of firms which are so different from anyone else.

4.1.2 | Deciding which outcomes to look at (Y)

Theoretical discussions of experimental methodology typically write down a single outcome Y of interest. But in practice there are often many possible outcomes of interest, and the same study can be well powered to detect impacts on some outcomes while having very little power to detect impacts on other outcomes.

A first point to note is that power will typically be greater for measuring outcomes closer in the causal chain to the intervention, something I call the funnel of attribution in McKenzie (2018). For example, in our Colombian export programme example, a possible causal chain is that the consulting intervention helps firms improve their management practices. Management practices are then one component that helps determine firm productivity. Firm productivity in turn is one component that determines whether or not firms try to export, and some of those firms that try to export are successful and able to sell. The further one moves along the chain, the more additional shocks and other factors may also be affecting the outcome, and the harder it may become to isolate the effect of interest. Another way of considering this is to think of it in multiplicative terms: the effect of consulting on exports equals the effect of consulting on management practices times the effect of management practices on productivity times the effect of productivity on trying to export times the effect of trying to export on export sales. If impacts are a few percentage points or 0.1 to 0.2 standard deviations in each step, the impact on the final product will be much less. This highlights the importance of measuring the most proximate and immediate outcomes.[8]

Second, the way we define an outcome can matter for the power we have to detect an effect. For example, consider outcomes related to export value in our example. We could consider the level of exports, the level of exports winsorised at some level (e.g. the 95th percentile), some transform such as log exports (although there are many zeros here), or a binary variable (e.g. exports exceeding a certain threshold such as $100,000 per year). When the outcome is heavily skewed, as is often the case for financial outcomes such as consumption, income, profits and exports, then we may have very low power to detect reasonable effect sizes when defining Y as the raw outcome, but more power or

[6] This was the approach taken by Iacovone, McKenzie and Meager (2025). Since their main outcomes of interest were around exports and export practices, they defined a stratum of 19 firms that were outliers in these variables.

[7] Policymakers may also be more interested in the TOT (treatment effect on the treated) in such cases, with the idea that scale-up may then focus on the more interested participants.

[8] These intermediate outcomes may be of less academic and policy interest than final outcomes, but it may be that a given study is not able to inform us of impacts on final outcomes. Then if we see an impact on intermediate outcomes, and have other studies that show that improvements in these intermediate outcomes typically lead to improvements in the final outcome, the intervention may still be considered a success. Here, the intermediate outcomes could be used to form a surrogate index for the final outcome (Athey et al., 2025).

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smaller MDEs for transforms which are less sensitive to the top tail. For example, consider the power to detect a 50 per cent increase in exporting. In raw levels, the mean is 338 and standard deviation (s.d.) 1170, so power is 17.5 per cent. Winsorising levels at the 95th percentile reduces the mean to 220 and s.d. to 452, so power is 40.6 per cent. Defining a binary outcome of more than $100,000 in exports has mean of 0.30 and s.d. of 0.46, and the power to detect a 50 per cent increase in this (to 45 per cent exporting) is 63.5 per cent.[9] One potential concern with changing the estimand here by winsorising or creating a binary variable is that it could miss detecting impacts of treatments that mostly work in the top tail. But one could also define a binary outcome with a higher threshold (e.g. being above the 90th percentile of the control group) to examine top tail outcome impacts.

A third way of changing the outcome being considered is to use multiple post-treatment measures, and then define the outcome as the average effect on the outcome pooled over multiple time periods. McKenzie (2012) discusses how collecting more follow-up rounds can improve power for a given cross-sectional sample size *N*. The gain in power from multiple follow-up rounds is greater when outcomes are not strongly autocorrelated in the follow-up rounds. Intuitively, this is because each follow-up then provides additional information about the treatment effect, whereas if outcomes are strongly autocorrelated, once you observe outcomes once, the subsequent rounds will be very similar and not tell you anything new. Weak autocorrelation is often the case for episodic health conditions and for volatile economic outcomes such as business profits or household consumption. For example, de Mel, McKenzie and Woodruff (2008) randomly give small grants to Sri Lankan microenterprises, and then collect eight quarterly follow-up rounds of business profits. Business profits are very noisy, and power would have been low in their sample to detect impacts on profits in a single follow-up round (e.g. impacts six months after treatment). But pooling together the eight follow-up rounds and redefining the estimand as the average impact over the first two years post-treatment averages out a lot of the idiosyncratic noise in firm profits and increases power – effectively reducing σ^2 in equation (6).

A couple of additional considerations can arise when conducting cluster-randomised studies. The first is that the intra-cluster correlation (ICC) may differ substantially across outcomes. For example, ICCs tend to be a lot higher for education outcomes such as test scores than for outcomes such as consumption or business profits. Kelcey, Shen and Spybrook (2016) find ICCs for reading and mathematics test scores in a range of sub-Saharan African countries to average 0.27–0.33. In contrast, ICCs for consumption, income and profits are much more commonly around 0.05. All else equal, the same clustered experiment will have more power for detecting impacts on outcomes with lower ICCs. Note that this also means that the choice of the optimal allocation of the number of clusters versus number of units per cluster will vary with outcome. Second, as noted above, the presence of unequally sized clusters can also reduce power. Imbens (2020) notes that there are two different estimands that then may be of interest: the population average effect (e.g. the impact for the average student or the average household) and the cluster average effect (e.g. the impact for a student in the average school or for a household in the average village). Power may be much lower for the former if a few large clusters dominate than for the latter which reweights each observation by the inverse of its cluster size. Cooper and McKenzie (2020) illustrate this with a cash transfer programme for workers, where most workers work in small firms but there are a few firms (clusters) that have 50 or more workers. They show the MDE for the cluster average effect (impact on the worker in the average firm) is 0.08 s.d., less than one-third of that for the population average effect (impact for the average worker). Aiming for an equal number of units per cluster, or redefining the estimand of interest to be the cluster average effect, is therefore recommended for boosting power.

[9] This uses the formula for a continuous outcome, assuming the variance does not change. Using this formula and having the standard deviation change to 0.497 as the proportion changes, or using the binary outcome formula, will give a power of 59.7 per cent or 59.2 per cent respectively.

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4.2 | The choice of treatment(s)

The main two design choices that affect statistical power are how many treatments to include in the experiment and how intense each treatment is. Researchers often would like to test different variants of a treatment. Sometimes, this is because there are design choices that are unclear (e.g. whether a cash transfer should be for $200 or $500, in a lump sum or instalments, paid by mobile transfer or given in kind, etc.). Other times, this is because there is a bundled treatment (e.g. training, an asset transfer, health counselling, community meetings and some cash), and researchers and policymakers would like to know whether all these components are required, or whether one or two are generating most of the impact. Finally, researchers may wish to isolate specific economic mechanisms by delivering treatments that act on only one mechanism at once.

The concern is that this can often result in many treatment arms, with a smaller sample size, and consequently lower power, for detecting the impact of any given treatment. Moreover, if authors then wish to correct for multiple hypothesis testing across treatments, this further reduces power for a given sample size (see, for example, McConnell and Vera-Hernández (2025)). One solution has been to cross-randomise treatments, but Muralidharan, Romero and Wüthrich (2025) show that these factorial designs tend to be underpowered to detect interactions between treatments, and the main effects then should be interpreted as effects on a population that has been partially exposed to another treatment (changing A, and hence the estimand). Adaptive experiments (e.g. Kasy and Sautmann, 2021) offer one approach for starting with many treatment arms and pruning them down to the most promising, but require the outcome to be realised relatively quickly and larger sample sizes than is the case in many field experiments.

Given these issues, my advice is to err on the side of fewer treatments, and making treatments as intensive and powerful as possible in a first study. Then if the desired impacts are realised, follow-on work can attempt to unbundle the treatment or test less intensive and cheaper ways of delivering it. If multiple treatment arms are included, constructing arms that the researcher feels comfortable pooling together can be helpful. For example, Fafchamps et al. (2014) offer cash and in-kind grants to small businesses, and increase power by pooling these to measure the impact of 'a' grant.

4.3 | Design choices to reduce the variance and the ICC

4.3.1 | Design choices that affect the residual variance (σ^2)

It is easier to detect a difference in outcomes between the treatment and control groups the more similar these two groups are to one another in the absence of treatment. While randomisation ensures the two groups have similar characteristics in expectation, they can be made more similar by using methods such as stratification or pairwise matching in the random assignment. Bruhn and McKenzie (2009) discuss these choices of method with individual-level randomisation, and provide simulations to show the gains in power with different types of outcomes and sample sizes. The main gains in power come in small samples (e.g. $N \leq 100$), in cases where the outcome of interest is highly persistent and/or can be well predicted by a few baseline characteristics. For example, stratifying or matching on baseline test scores in education tends to improve power a lot for examining test scores as a follow-up outcome. In contrast, when outcomes are harder to predict and less persistent, such as consumption or the incomes of the self-employed, the power gains from stratifying or matching are much less. Bruhn and McKenzie (2009) note that researchers should control for the randomisation strata or matched pair dummies in regression (2).

More recent work by Bai (2022) shows that the matched pair design is the optimal form of stratified randomisation to maximise statistical precision for estimating the average treatment effect (or ITT in our setting above). However, this assumes that there is only a single outcome of interest, and once one starts considering multiple outcomes of interest, the optimal assignment for one outcome will differ

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from that for another. In McKenzie (2022), I note that there can also be issues with variance estimation and attrition that can arise with matched pairs, and that matched quadruplets can deliver most of the gains of matched pairs while being a little less sensitive to these issues. My recommended approach for individual-level randomisation is then the following steps:

Step 1: Decide if there are one or two key variables that you want to stratify on. This could be for reasons other than maximising balance on the main outcome of interest, such as thinking that there could be treatment heterogeneity by gender or geographic location. Use this to form a relatively small number of strata.

Step 2a: If there are only a couple of main outcomes of interest, and they are in the same family (e.g. profits and sales for a firm experiment, or maths test score and English test score in an education experiment), then form an aggregate index of these outcomes and sort units according to this index. Then form quadruplets within each stratum. For example, in the Colombian export example, we stratified by firm size and whether firms were exporting at baseline, and then within each of these strata, we formed matched quadruplets according to an index of export management practices.

Step 2b: If there are many different outcomes of interest in multiple domains, or if the outcome of interest is not available, or if it is the same for everyone (e.g. everyone in a training programme may be unemployed at the start of the study), then construct the Mahalanobis distance between units on a set of covariates that you think are likely to predict the main outcomes well. Then form matched quadruplets based on these distances. Note that this can be computationally intensive, especially as the sample size gets larger, and I think it is usually only worth considering when you have a very small number of units or clusters.

The effective sample size is typically smaller in clustered experiments due to the design effect in equations (7) and (8), making the power gains from stratifying or using matched pairs potentially more important in clustered experiments. Imai, King and Nall (2009) suggest also stratifying on cluster size when clusters differ in size, as well as covariates predictive of the outcome of interest. However, there are a couple of factors that reduce how much power researchers should expect to gain. The first is that stratification or matching is done on cluster-level covariates, not unit-level covariates. Often these covariates may differ a lot more within clusters than across clusters, and so by the time one averages them up to the cluster level, the cluster-level average does not predict very much of the follow-up outcome. For example, an individual's baseline test scores may be highly predictive of their test score one year from now. But the stratum dummy for being in a school with a high average baseline test score will typically be much less predictive. Second, de Chaisemartin and Ramirez-Cuellar (2024) show that analysis of clustered experiments needs to cluster standard errors at the level of the matched pair or small strata for tests to have the correct size, and then one can also run into issues of inference with a small number of clusters. A less conservative variance estimator than the OLS clustered standard error, such as the between-strata robust variance of Imbens and Rubin (2015), may improve power here. However, this is an area where the literature differs in assumptions on superpopulations versus fixed populations, homogeneous versus heterogeneous treatment effects, and asymptotic versus small sample results. I therefore currently advise researchers to use simulations to help guide their level of stratifying in cluster experiments, and to estimate power since standard formulae may not give correct results with stratification or small samples.

4.3.2 | Choices that affect the intra-cluster correlation (ρ)

The ICC can also be changed by researchers choosing more or less homogeneous entities to serve as clusters. Often, the clusters will be pre-determined, such as schools, villages or markets, but then researchers may be able to affect the ICC by which set of schools or villages they decide to include in their study. In other settings, it is the researchers themselves forming the clusters – for example,

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grouping firms into networking groups or students into activity groups. In both cases, choosing clusters in a way that the individual units are not highly sorted across clusters, so that ρ is small, will help improve power.

4.4 | Tips for power calculations in grant proposals, pre-analysis plans and registered reports

Funders and referees want to see that a prospective experiment being designed will have enough power to detect treatment effects of interest. The suggestions in this paper can help researchers design more powerful experiments. But it is also worth noting some common areas where proposals often fall short in providing power calculations. To my mind, one of the most frequent issues is researchers only providing an MDE in terms of standard deviations. If we set power to 0.8 and size to 0.05, and allocate half of the sample to treatment, then equation (5) reduces to

$$MDE = \frac{5.6}{\sqrt{N}}\sigma \tag{9}$$

and so one can easily calculate the MDE for a given sample size, with researchers then sometimes using cut-off rules for this term, sometimes called Cohen's d, in which 0.2 s.d. is considered a small effect, 0.5 s.d. a medium effect, etc. However, standard deviations are often not an economically meaningful or useful unit of measurement for effect sizes, and should be avoided when natural units such as percentage points, percentages or absolute monetary amounts are available. For example, consider our Colombian export example. The mean of exports was \$338,000 and standard deviation \$1.17 million. A 'small' 0.2 standard deviation MDE would then be equivalent to a \$234,000 increase in exports, or 69 per cent increase on the baseline mean. But if we had winsorised at the 95th percentile, a 0.2 standard deviation increase would be a \$90,000 increase, or 41 per cent increase in the mean. This problem of the standard deviation changing dramatically with the sample and not being interpretable as an absolute measure of impact is also shown for test scores by Singh (2015), who notes how dispersion can vary substantially with sample and aggregation choices. Instead, researchers should express MDEs in percentage point terms for binary outcomes (e.g. what is the percentage point increase in the probability of employment we have power to detect from this jobs programme?) and in units such as dollars or percentages for continuous outcomes such as consumption, incomes and profits.

Power calculations should clearly state assumptions about take-up rates and compliance, attrition, ICCs and other parameters that will determine power. They should also provide sufficient detail about the status quo and treatment to allow the reader to understand what is being changed, and attempt to benchmark the MDE against cost–benefit thresholds and/or the existing literature to show why the MDE is at a level that would be interesting to detect. As noted above, power will vary depending on outcome, and so providing calculations for several key outcomes of interest is desirable. Finally, in studies that aim to compare multiple treatments or look at heterogeneity in treatments, the power calculations should also consider the more stringent sample sizes needed for such comparisons. For example, Gelman (2018) argues that under reasonable assumptions about main effects and interactions, you need 16 times the sample size to estimate an interaction than you need for estimating a main effect.

5 | IMPROVING POWER AT THE IMPLEMENTATION STAGE

The power of an experiment can be further influenced by researcher effort and actions during the implementation phase of treatment. This is particularly the case for field experiments that rely on surveys to collect outcome data.

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5.1 | Boosting take-up and reducing non-compliance

It is easier to detect a large treatment effect than a small treatment effect. From equation (1), we see that the ITT is the difference in mean outcomes between the treatment group and the control. This gap will be smaller if many of those assigned to treatment do not take it up, and if some of those assigned to the control end up receiving treatment. Let π_T and π_C be the take-up rates in the treatment and control groups respectively. If the treatment effect for those taking up the treatment (TOT) is a constant γ, then from equation (5) we see that the sample size needed to detect effect size $\gamma(\pi_T - \pi_C)$ will be proportional to $(\pi_T - \pi_C)^{-2}$, i.e. to the inverse square of the difference in take-up rates. So a take-up rate difference of 25 percentage points would require four times the sample size to have the same power as when there is 100 per cent compliance. Ibarra et al. (2021) show that this loss of power is less if treatment effects are heterogeneous and treated individuals sort into take-up based on their individual treatment effects: if there is a subgroup that actually have negative treatment effects, trying to push for 100 per cent take-up could reduce power compared with having 80–90 per cent take-up. But given the difficulty individuals typically have in knowing their likely treatment effects, typically power will be maximised when take-up is boosted among the treatment group to as close to 100 per cent as possible, and when none of the control group receives treatment.

Take-up can be influenced in the design phase through screening individuals on their interest in the treatment, and defining the population A to be those more likely to take it up. In particular, this suggests a preference for oversubscription designs over encouragement designs. For example, consider two approaches for conducting an experiment to measure the impact of a new microfinance savings product, where mean savings in the control group are 100 with an s.d. of 50. In an encouragement design, a list of 1000 microfinance clients might be divided into two equal groups, with the 500 in the treatment group invited to use the new savings product. But since this is not screened on interest, we might expect perhaps only 30 per cent to take up the treatment. In an oversubscription design, clients would first be asked whether they were interested, and so perhaps 1000 clients get screened to 250 who say they are interested. Then if 125 get randomly assigned to treatment, we might expect 90 per cent to take up the product given they had already expressed interest. Then what is the power of these two approaches to detect a 25 per cent TOT? In the encouragement design with the sample of 1000, the ITT is $0.3 \times 25 = 7.5$ per cent and power is 66.0 per cent. In the oversubscription design, with a sample of 250, the ITT is $0.9 \times 25 = 22.5$ per cent and power is 94.5 per cent.[10] That is, power is substantially higher despite having only one-quarter of the sample, because we have screened out mostly individuals who were unlikely to take up treatment.

Conditional on the given treatment sample, take-up can be increased by efforts to ensure the treatment group is well informed about the programme and reminded of what they need to do to take up treatment and that it is easy for them to take part in treatment, along with potentially offering incentives to participate. Here we can distinguish between efficacy trials, in which the aim is to deliver the treatment with high fidelity to as many in the treatment group as possible, to learn whether the treatment works; and effectiveness trials, where the goal may be to learn the effect of treatment as implemented by a government or NGO without researcher involvement, in which costly efforts to increase take-up are not implemented. This comes back to deciding what the estimand of interest is: the impact of a treatment implemented with researcher supervision on a population screened for interest is different from the impact of one implemented at scale or on a random sample with government implementers.

A common threat to non-compliance is that the implementer wants to treat a certain number of units, and then if there is insufficient take-up from the treatment group, they will invite members of the control group to make up the difference. For example, a training programme provider may have a contract to train 500 jobseekers, and if only 350 of the 500 assigned to treatment show up, they

[10] In Stata: sampsi 100 107.5, n1(500) n2(500) sd1(50) versus sampsi 100 122.5, n1(125) n2(125) sd1(50).

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may try to find 150 in the control group to meet their quota. One way researchers can anticipate and mitigate this risk is through the use of a waiting list: randomly assigning units to treatment, control and a waiting list. For example, Hirschleifer et al. (2016) evaluated the impact of a government vocational training programme for the unemployed in Turkey. They chose 2.2 times as many applicants as there were slots q in the classes, and put q in treatment, q in control and $0.2q$ into a waiting list that the provider could contact in any order and use to meet their quotas. This waiting list was then not part of the experimental evaluation. An alternative is to randomly reassign units from a waiting list to treatment until the quota is filled, and then use a weighted estimator that takes account of this procedure (de Chaisemartin and Behagel, 2020).

5.2 | Reducing attrition

The sample size N available for estimating treatment effects, and hence statistical power, gets reduced if outcome data are missing for some units. This can happen with administrative data if they are only collected or can only be accessed for a subset of observations. Researchers may have little control over this form of attrition.[11] In contrast, with survey data, the time and effort devoted to tracking down individuals and getting them to respond to surveys can have a large impact on attrition. Persistence in using multiple callbacks, and then efforts to provide alternative ways for individuals to respond, can greatly boost response rates. For example, McKenzie (2015) shows that the common approach of just doing three callbacks would have yielded a response rate of 51 per cent in the treatment group and 48 per cent in the control group in an in-person survey of Nigerian entrepreneurs. However, continuing to call back until the person was found or refused boosted these response rates to 67–74 per cent, and then adding online and phone survey options increased response rates further to 83–89 per cent. Thomas et al. (2012) provide further guidelines and experience for reducing attrition. Of course, attrition can affect bias as well as power, and so efforts should be made to reduce differential attrition in addition to overall levels.

5.3 | Reducing σ^2 through careful attention to measurement

Power is higher the smaller is the residual variance. One reason that this variance can be high is from measurement error in the outcome Y. Careful attention to measurement and reducing measurement error can potentially increase power substantially. This can include embedding consistency checks in the survey so that unusual values get checked (e.g. Fafchamps et al., 2012), especially when dealing with currency units with lots of zeros (McKenzie, 2014), and using anchoring and triangulation (e.g. Anderson, Lazicky and Zia, 2021). For outcomes such as knowledge, behaviours and practices, asking multiple questions about the same concept and averaging them together can reduce noise. Administrative data, when available, may also improve precision, but may also have their own measurement errors. From equation (5), we see that the MDE is proportional to σ, and so an x per cent reduction in the residual variance from better measurement will reduce the minimum detectable effect size by the same proportion, or equivalently boost power to detect a given MDE.

6 | IMPROVING POWER AT THE ANALYSIS STAGE

Having designed and implemented an experiment, there are still further choices researchers can make at the analysis stage to improve statistical power and learn as much as possible from the experiment.

[11] If matching to administrative data requires using ID numbers collected on application forms or surveys, researchers can implement checks to ensure these are entered correctly in the first place.

These include changing the test statistic or estimand, the choice of control variables, and using outside information in a Bayesian approach. While these are implemented at the analysis stage, they should be pre-specified when possible, but otherwise could be used as additional exploratory analysis.

6.1 | Changing the test statistic or estimand

Standard power analysis is based on the two-sided test in equation (4), that the ITT is zero. One-sided tests can increase the power to detect impacts in one direction. This could be useful for policy decisions if, for example, a policy will be introduced if the estimated net benefit is positive. However, for economic research, I think we should stick with two-sided tests, since there are many examples of policies that have the opposite impact to the one intended. For example, in the Colombian export example, Iacovone, McKenzie and Meager (2025) find that some export outcomes fell, rather than increased, after the consulting intervention. Their qualitative work found that consultants told some firms they would not be competitive in international markets and to focus more on their domestic market. As another example, Kelley, Ksoll and Magruder (2024) find that a programme to try to increase employment of vocational training graduates in India by uploading them to a digital platform and sending them text alerts about job openings actually reduced the probability of employment, by increasing reservation wages. We want tests that can detect these negative impacts, rather than just being able to rule out a significant positive impact.

When take-up of a treatment is low, or non-compliance is high in the control group so that many of them are treated, researchers may be able to precisely estimate that the ITT is close to zero. But the standard instrumental variables (IV) estimator of the treatment effect on the treated (TOT) may not be very precise. For example, Ibarra, McKenzie and Ruiz-Ortega (2021) conducted an experiment with over 100,000 credit card clients in Mexico, in which treatment groups were randomly assigned to financial education workshops or coaching. Take-up rates were only 0.8 per cent for the workshops and 6.8 per cent for the coaching, and the estimated ITTs were close to zero, with small standard errors. However, with this low take-up, the TOTs had wide confidence intervals: for example, the impact on paying more than the minimum credit card payment had a 95 per cent confidence interval of –25 percentage points to +18 percentage points, representing an almost halving of the control mean at the low end to a one-third increase at the high end. The authors tried to look at impacts on those who received treatment by matching them to control individuals and using difference-in-differences, getting much more precise impacts. Coussens and Spiess (2021) provide a method that does not require using non-experimental methods, but that instead uses machine learning to estimate compliance with treatment as a function of baseline covariates, and then constructs a weighted IV estimator that puts more weight on the units that are more likely to comply. If compliance can be predicted by baseline covariates, this can yield a more precise estimate, although of a different estimand which can be interpreted as a convex combination of conditional local average treatment effects.

When there are multiple outcomes, then if these outcomes are all in the same family or domain, researchers often combine them into a single index. These indices typically weight each outcome equally or use inverse covariance weighting. Anderson and Magruder (2023) provide an alternative index, which is constructed via a data-driven sample-splitting approach to maximise power for detecting mean treatment effects on a subset of indicators in the data by putting more weight on indicators for which there is a larger mean difference. This can be more powerful for detecting whether the treatment had *some* effect in a particular domain, but interpretation of the index and magnitude is more difficult. Another test with multiple variables is the omnibus test of Young (2019), which is a test of overall experimental significance. Here, the null hypothesis is that no treatment had any effect on any outcome for any unit. There may then be occasions where we can learn that the treatment had some effect, even if the experiment is underpowered for detecting impacts on any specific outcome.

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6.2 | Choice of control variables

The residual variance σ^2 reflects how much individual outcomes vary around the treatment and control means. Power can be increased by adding additional control variables to the OLS equation (2) that help predict the outcome Y, making it easier to detect any remaining difference that is coming from treatment. The most important of such controls is typically the lagged outcome variable. Adding one or more lags of the outcome in an ANCOVA regression can result in large improvements in power, with the gains higher the more autocorrelated is the outcome of interest (McKenzie, 2012). The autocorrelation tends to be very high for test scores, and lower for outcomes such as consumption and profits. However, sometimes the outcome of interest is not available pre-treatment, or may be the same for everyone at baseline (e.g. a group of non-exporters all have zero exports and jobseekers all have zero employment). The question is then whether there can be much gain in power from adding covariates other than the lagged outcome variable (and any randomisation strata).

The post-double selection Lasso (PDS Lasso) estimator of Belloni, Chernozhukov and Hansen (2014) has grown in popularity as a way for researchers to choose which additional control variables from a large potential list to include as additional controls in their field experiments. This method selects variables that are strongly predictive of the outcome or of treatment. Cilliers, Elashmawy and McKenzie (2024) reanalyse published field experiment papers and provide lessons for both the usage of this method and how much gain in power researchers should expect. They find that in practice this method tends to choose very few control variables, with a mean of 3.6 and median of 2, and that on average it reduces standard errors by less than 1 per cent compared with just ANCOVA. Even when no baseline outcome is available for ANCOVA, the median standard error after using PDS Lasso is only 0.99 of that just using strata controls, although the 25th percentile is 0.94 of the OLS standard error, and there are a small proportion of outcomes that see larger reductions. Therefore the average study is unlikely to see much of a gain in power from using this method, reflecting how hard it often can be to predict future outcomes (e.g. McKenzie and Sansone, 2019), but there will be cases where power can be boosted by selecting a few key additional controls.

6.3 | Incorporating outside information through Bayesian analysis with informative priors

A final way to increase the precision of our estimates is through incorporating outside information, beyond the data collected from the experiment itself. Iacovone, McKenzie and Meager (2025) illustrate how to do this in the Colombian export example through incorporating informative priors in a Bayesian impact evaluation. With diffuse, or non-informative, priors, the 95 per cent posterior intervals from a Bayesian impact evaluation correspond to the frequentist 95 per cent confidence intervals. However, this ignores information about the likely effects of the treatment that comes from previous research, economic theory and policy expertise. A prior distribution of the likely impact of a treatment can be calculated based on previous literature, or from eliciting these priors from academics, policymakers or programme participants. For outcomes in which these priors are in line with the results of the experiment, incorporating this information results in posterior intervals that can be considerably narrower than frequentist confidence intervals. For example, Iacovone et al. (2025) find that the posterior intervals for whether firms increased export variety were 55–68 per cent as wide as the frequentist intervals. When the experimental results differ from the priors, then the certainty with which these priors were held and the degree of precision in the experimental results determine how much the priors should be updated in light of the experiment. This approach can be particularly useful when sample sizes are fixed by policy constraints, but researchers could also consider a Bayesian version of power analysis if sample size is a choice (Goldfeld, 2021).

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7 | CONCLUSIONS

Many experiments in the economics literature appear to be underpowered to detect policy-relevant impacts for key outcomes. Often the sample size for experiments is fixed by the number of eligible applicants to a programme or by budget constraints. This paper shows that even with a fixed sample size N, there are many researcher choices at the design, implementation and analysis stages that can influence statistical power and ensure that we learn as much as possible from a given study. Box 1 summarises these choices. A consequence is that it does not make sense to talk of 'the' power of an experiment: experiments may be well powered for some estimands and outcomes and not others, and there is much that researchers can do to improve power.

BOX 1 Summary of methods researchers can use to improve power

1. At the design stage

Change the estimand: eliminate units that are likely to attrit, that are likely to not comply or that are outliers.

Choose outcomes that you have more power to detect: measure outcomes closer to the intervention in the causal chain; consider winsorised or binary versions of continuous outcomes; use multiple rounds of post-treatment measurement. In clustered studies, consider outcomes with low ICCs, and ensure cluster sizes are not too unequal.

Be judicious in how many and which treatments are used: do not add too many arms; make treatments more intensive or in a way that can be pooled.

Randomise in a way to reduce the variance: use stratified or matched quadruplet random assignments.

2. At the implementation stage

Boost take-up and reduce non-compliance: screen on interest; use reminders and other approaches to help make treatment easy to receive; use a waiting list to reduce the chance of the control group getting treated.

Reduce attrition: collect multiple ways of contacting participants; make multiple survey attempts; look for other data sources such as administrative or web data.

Reduce the variance through better measurement: incorporate data checks and triangulation approaches; combine multiple measures; be careful with issues such as numbers containing many zeros.

3. At the analysis stage

Consider alternate test statistics: there may be more power to detect impacts on a subgroup of units more likely to take up treatment, for indices that put more weight on indicators that show changes, and through using tests that test whether there is any effect (without measuring magnitudes).

Choose control variables to further reduce variation: the lagged dependent variable will typically be the most important, but other controls that can predict the outcome can also improve power.

Incorporate outside information through Bayesian analysis: use informative priors from the literature or elicited from experts or participants.

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CONFLICT OF INTEREST STATEMENT

There are no conflicts of interest.

DATA AVAILABILITY STATEMENT

Replication code and data are available in the World Bank's Reproducible Research Repository https://reproducibility.worldbank.org/catalog/305.

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